Alexander Akerman

Software Engineer | AI Platforms | Computer Vision | Backend Systems

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Professional Summary

Software engineer with 15 years at eBay, serving as lead engineer for multiple production backend, AI, and media-processing systems. Combines architectural decision-making and hands-on implementation with production operations and cross-team delivery.

Professional Experience

eBay **2011-Present**

Software Engineer | AI Platforms and Computer Vision **2019-Present**

AI Platforms and Production Services

- Developed a production live-commerce product identification system combining video, speech, and auction change data capture (CDC) events using Flink, Kafka, and Gemini Live API.
- Designed and implemented retrieval-augmented generation (RAG) for a production agent serving external eBay developers, including embedding storage in a vector database, query embedding, and retrieval of official developer documentation.
- Owned architecture, implementation, deployment, and on-call operations for a production image embedding pipeline handling approximately 2,400 images per second at peak. Coordinated design and capacity requirements across AI, partner engineering, risk, and compliance teams.
- Built a Spring Boot starter and tracing tools for an AI agent framework adopted by multiple production agents.

Computer Vision and Model Serving

- Led development of trading-card image matching and built automated index generation, reducing delays before new cards became available for matching.
- Integrated machine learning models into production serving infrastructure, porting Python preprocessing and post-processing to Java.
- Integrated SIFT and FFmpeg libraries with Java through JNI for production image matching, media inspection, and frame extraction.

Media Processing and Moderation

- Led modernization of a video moderation pipeline handling approximately 20,000 videos per day, with capacity for approximately 80,000 per day. Completed migration ahead of schedule.
- Built a production image classification and OCR pipeline handling approximately 4,500 image uploads per second using Kotlin, Spring Boot, Kafka, and OpenCV.

Software Engineer **2014-2019**

Metadata Tools

- Developed and maintained a full-stack metadata application for internal users using React, Java, Spring, and Neo4j.

Software Engineer **2011-2014**

Data Ingestion and Processing

- Maintained production-critical vendor data ingestion and ETL (extract, transform, load) infrastructure using Perl, Java, and Linux.

Technical Skills

Languages: Java, Kotlin, Python, JavaScript, Perl

Backend and Infrastructure: Spring Boot, Kafka, Flink, REST APIs, Docker, Kubernetes, Avro, Amazon S3

AI and Computer Vision: NVIDIA Triton Inference Server, OpenCV, OCR, FFmpeg, SIFT, Gemini Live API

Platforms and Tools: Java Native Interface (JNI), MongoDB, Neo4j, React, Linux, Maven, CMake, Micrometer